

By summing up these integrals, some terms cancel out and we get

$$\begin{aligned}\int_P x^\top v \, dx &= \sum_{i=1}^m \frac{v_2(q_{i,2} + q_{i-1,2}) + v_1(q_{i,1} + q_{i-1,1})}{6} (q_{i-1,1}q_{i,2} - q_{i,1}q_{i-1,2}) \\ &= \sum_{i=1}^m \frac{v^\top (q_{i-1} + q_i)}{3} \frac{q_{i-1,1}q_{i,2} - q_{i,1}q_{i-1,2}}{2}.\end{aligned}$$

For the calculation of W , we sum up these integrals with $P = D_i(w)$ and $v = a_i$ for all $i \in \{1, \dots, n\}$. For practical reasons we write this in terms of the edges, where E denotes all edges. Let $e \in E$ be an edge with vertices q_1^e, q_2^e . Let $a_l, a_r \in \mathbb{R}^2$ be the vectors a_i on the left/right side of e . We set $a_l = 0$ or $a_r = 0$ for the non-existing side of boundary edges. Previously, every edge was considered twice, thus we can write

$$\sum_{i=1}^n \int_{D_i} x^\top a_i \, dx = \frac{1}{3} \sum_{e \in E} \left((a_l - a_r)^\top (q_1^e + q_2^e) \right) \frac{q_{1,1}^e q_{2,2}^e - q_{2,1}^e q_{1,2}^e}{2}.$$

For an inner edge, the addend can be rewritten as

$$2(w_l - w_r) \frac{q_{1,1}^e q_{2,2}^e - q_{2,1}^e q_{1,2}^e}{2}.$$

4 Numerical algorithms

In this section, we address numerical algorithms for the solution of (Pw).

4.1 Fixed-point iteration

As an optimality condition for (Pw) we derived $r(w) = 0$, cf. Theorem 2.6. This can be transformed into the fixed-point equation $w = w + \tau r(w)$ with $\tau > 0$. A possible algorithm for the solution is the fixed-point iteration $w_{k+1} := w_k + \tau_k r(w_k)$ with appropriate step sizes $\tau_k > 0$. It is not clear, how to choose these step sizes. We will see in Lemma 4.2 below, that the direction $r(w_k)$ is a descent direction for J . However, since J is not C^1 , Armijo step sizes could become arbitrarily small and the usual convergence proof does not work. Similarly, constant (but small) step sizes might not give a decrease of J .

Instead, we can view the above iteration scheme as a discretization of an ODE. Indeed, the above update formula yields $(w_{k+1} - w_k)/\tau_k = r(w_k)$. This is an explicit Euler scheme for the ODE

$$\begin{aligned}w'(t) &= r(w(t)), \\ w(0) &= w_0.\end{aligned}\tag{ODE(w)}$$

We state some simple properties for (ODE(w)).

Lemma 4.1. *The following properties hold.*

- (i) *The function r is globally Lipschitz continuous with constant $L > 0$.*